VISVESVARAYA TECHNOLOGICAL UNIVERSITY

JNANA SANGAM, BELAGAVI – 590 018

A PROJECT SYNOPSIS
ON

"Predictive Analytics on Twitter Sentiment Data"

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By

MOHAMMED ASIF
(1KN22AI025)

MOHAMMED SIDDIQ
(1KN22AI030)

Under the guidance of
Mrs.SALMA HUSSAINNAIK
Assistant Professor, Dept. of AIML Engg.
K.N.S.I.T, Bangalore.

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DEPARTMENT OF ARTIFICIAL INTELLIGENCE & MACHINE LEARNING

KNS INSTITUTE OF TECHNOLOGY

THIRUMENAHALLI, BANGALORE – 560064

ABSTRACT

Our project implements sentiment analysis by collecting textual data through web scraping and applying preprocessing techniques like tokenization, stemming, and TF-IDF vectorization. Various machine learning models were trained, with the Random Forest Classifier achieving the highest accuracy of 96%. Challenges include sarcasm detection, multilingual processing, and emoticon handling. Future work aims to enhance model capability for detecting satire and sarcasm automatically.Our analysis highlights the importance of data quality and feature engineering in improving model performance. Implementing deep learning techniques like LSTMs or transformers could further enhance sentiment classification. Integrating real-time sentiment tracking may also provide valuable insights for businesses and social media monitoring.

INTRODUCTION

In today's digital era, understanding customer sentiment is crucial for businesses to enhance satisfaction and improve services. Sentiment analysis, a branch of NLP, helps classify emotions—positive, negative, or neutral—within text data from reviews, social media, and feedback. This process involves data collection, pre-processing, feature extraction, and machine learning-based classification. Businesses use sentiment analysis to monitor brand perception, detect issues, and make data-driven decisions. However, challenges like sarcasm, irony, and multilingual text complicate accuracy. This synopsis explores the methods, tools, applications, and future directions of sentiment analysis in transforming customer feedback into insights.

LITERATURE SURVEY

1. **Foundational Work in Sentiment Analysis**

Authors: Pang et al. (2002), Liu (2012)

Summary: Pang et al. pioneered machine learning for sentiment classification using Naive Bayes, SVM, and Maximum Entropy. Liu (2012) expanded the field by focusing on subjectivity detection and opinion summarization, forming the basis for modern sentiment analysis techniques.

2. **Pre-Processing Techniques**

Authors: Bird et al. (2009), Jurafsky and Martin (2020)

Summary: Tokenization, stemming, and lemmatization are key for text normalization. Tools like NLTK (Bird et al.) and regex-based cleaning (Jurafsky & Martin) help remove noise, improving data quality for analysis.

3. **Feature Extraction Methods**

Authors: Mikolov et al. (2013), Pennington et al. (2014)

Summary: Traditional methods like BoW and TF-IDF convert text into numerical features. Advanced techniques like Word2Vec (Mikolov et al.) and GloVe (Pennington et al.) capture semantic relationships, enhancing model accuracy.

4. **Machine Learning Algorithms**

Summary: Decision Trees, Random Forests, SVM, and Logistic Regression are widely used for sentiment classification. Random Forests handle missing data well, while SVM and Logistic Regression perform strongly in binary and multi-class tasks.

5. **Challenges and Limitations**

Authors: Medhat et al. (2014)

Summary: Sarcasm, irony, and multilingual text complicate sentiment analysis. Rule-based systems lack scalability, leading to a shift toward automated machine learning approaches (Medhat et al.).

6. **Recent Trends and Future Directions**

Authors: Devlin et al. (2018)

Summary: Deep learning models like RNNs and BERT (Devlin et al.) have revolutionized sentiment analysis. Future research focuses on multilingual support, sarcasm detection, and real-time analysis systems.

PROJECT OVERVIEW

a. Problem Statement:

Sentiment analysis is crucial for businesses to understand customer emotions from online feedback, but traditional methods struggle with challenges like sarcasm, multilingual text, and scalability. This project aims to develop an automated sentiment analysis system using machine learning to classify text into positive, negative, or neutral sentiments, addressing these limitations and improving accuracy.

b. Objectives and Scope of the Project:

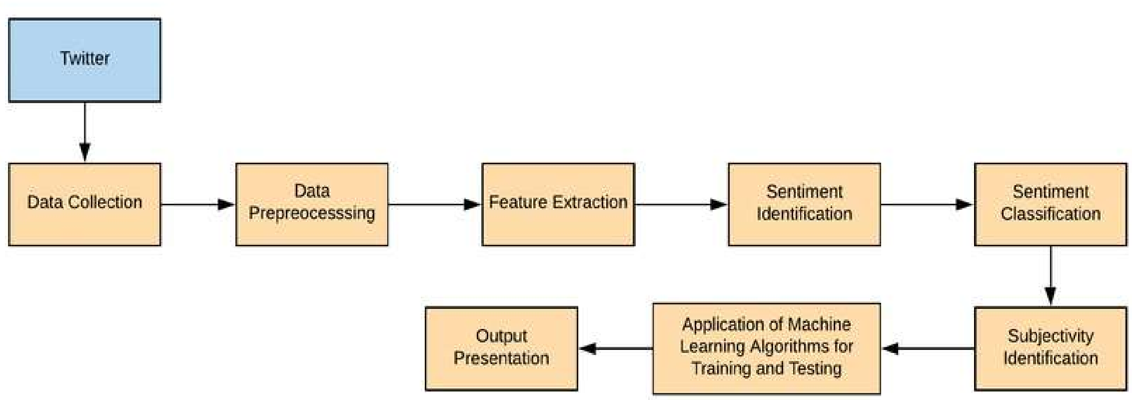
Objectives:

1. To pre-process and clean textual data (e.g., tweets, reviews) for analysis.
2. To implement and compare machine learning algorithms for sentiment classification.
3. To achieve high accuracy in predicting sentiment while handling challenges like sarcasm and multilingual text.

Scope:

The project focuses on analyzing customer feedback from social media platforms like Twitter. It includes data collection, pre-processing, feature extraction, model training, and evaluation using metrics like accuracy and F1-score.includes data collection, pre-processing, feature extraction, model training, and evaluation using metrics like accuracy and F1-score.

c. Block Diagram/ Architecture



d. Algorithm

1. **Decision Trees :** A flowchart-like model where each internal node represents a decision based on a feature, each branch an outcome, and each leaf node a class label (e.g., positive/negative sentiment). Simple to interpret but prone to overfitting without proper pruning.
2. **Random Forest Classifier:** An ensemble method that builds multiple decision trees and merges their predictions for improved accuracy and robustness.Handles missing data well and reduces over fitting compared to single decision trees.
3. **K-Nearest Neighbors (KNN):**A non-parametric algorithm that classifies data points based on the majority class among their k closest neighbors.Simple but computationally expensive for large datasets due to distance calculations.
4. **Logistic Regression:** A statistical model used for binary classification (e.g., positive/negative sentiment) by estimating probabilities using a logistic function.Efficient for linearly separable data but struggles with complex relationships.
5. **Support Vector Machine (SVM):** A supervised learning model that finds the optimal hyperplane to separate classes in high-dimensional space.Effective for both linear and non-linear classification using kernel tricks.

Software Requirements

Programming Language	**:** Python, NLP,LLM
Development Environment	**:** Google COLAB, Jupyter Notebook
System	**:** MicrosoftWindows10or11
Libraries	**:** NUMPY,PANDAS,MATPLOTLIB,SEABORN,SKLEARN

Hardware Requirements

Processor	**:** i5, i7, NVIDIA RTX (2GB,4GB)
Output Devices	**:** Monitor (LCD)
Input Devices	**:** Keyboard, Mouse
Hard Disk	**:** 512 OR ABOVE
RAM	**:** 16GB or above

METHODOLOGY

1. **Data Collection:** Extract textual data using web scraping and import datasets for analysis.
2. **Preprocessing:** Clean text by removing punctuation, tokenizing, stemming, lemmatizing, and vectorizing.
3. **Feature Extraction:** Convert text into numerical format using Count Vectorization and TF-IDF.
4. **Model Training:** Train machine learning models like Decision Trees, Random Forest, KNN, Logistic Regression, and SVM.
5. **Sentiment Classification:** Categorize text into positive, negative, or neutral sentiments.
6. **Result Analysis & Comparison:** Evaluate model accuracy and identify the best-performing classifier

TECHNOLOGY USED

1. **Python:** Primary programming language for data processing and model implementation.
2. **NLTK & TextBlob:** Libraries for natural language processing, tokenization, stemming, and sentiment analysis.
3. **Scikit-learn:** Machine learning library for implementing classifiers like Decision Trees, Random Forest, and SVM.
4. **Pandas & NumPy:** Data manipulation and numerical computing libraries.
5. **Matplotlib & Seaborn:** Visualization libraries for analyzing sentiment distribution and model performance.
6. **Jupyter Notebook:** Development environment for testing and evaluating the sentiment analysis models.

EXPECTED OUTCOME

1. **Accurate Sentiment Classification:** The model will classify text into positive, negative, or neutral sentiments with high accuracy.
2. **Best Performing Model Identification:** Among the implemented algorithms, Random Forest is expected to perform best based on accuracy evaluation.
3. **Real-Time Sentiment Analysis:** The system will provide quick and automated sentiment classification for large datasets.
4. **Improved Customer Insights:** Businesses can analyze customer opinions, reviews, and feedback efficiently.
5. **Scalability & Future Enhancements:** The model can be extended to detect sarcasm, multilingual sentiment, and emoji-based sentiment analysis.

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